

# Sub-Millijoule Intrusion Detection on a Commodity MCU Neural Processing Unit

Anonymous  
Double-blind review

**Abstract**—We deploy an intrusion detection system (IDS) classifier on the STM32N6570-DK, a commodity ARM Cortex-M55 MCU with the Neural-ART NPU rated at 600 GOPS INT8. Exploiting the approximate equivalence between single-timestep ( $T=1$ ) spiking neural network (SNN) inference and INT8 quantized ANN inference, we compile a lightweight MLP to the NPU without neuromorphic hardware. On NSL-KDD (5-class) and UNSW-NB15 (10-class) with 10 random seeds, the model achieves  $78.86 \pm 1.32\%$  and  $64.75 \pm 0.61\%$  overall accuracy, respectively. INT8 NPU inference completes in 0.29–0.46 ms ( $2.7\text{--}4.2\times$  faster than CPU-only), with an estimated energy of 44–69  $\mu\text{J}$ /inference and a Flash footprint of 120–138 KB. Compared with recent MCU-class deployments on the STM32F7 (31 ms, 7.86 mJ) and Raspberry Pi 3B+ (27 ms), our NPU path achieves  $59\text{--}107\times$  lower latency and  $114\text{--}179\times$  lower energy than the STM32F7. To our knowledge, this is the first publicly documented IDS deployment on an ARM Cortex-M NPU, and also the first non-vision workload reported on the Neural-ART accelerator.

**Index Terms**—intrusion detection, spiking neural network, neural processing unit, INT8 quantization, edge AI, STM32N6

## I. INTRODUCTION

Deep-learning-based intrusion detection at the IoT edge must operate under tight power and latency budgets. Recent MCU-class deployments achieve promising accuracy but at high energy cost: Chehade et al. reported 31 ms and 7.86 mJ per inference on an STM32F746G (Cortex-M7, no NPU) [1], and Diab et al. required 27 ms on a Raspberry Pi 3B+ [2]. At these latencies, CPU-only inference blocks all concurrent RTOS tasks for tens of milliseconds, making real-time packet processing difficult.

Commodity MCUs are beginning to integrate neural processing units (NPUs) that execute INT8 matrix operations at hundreds of GOPS while freeing the CPU for other tasks. A line of theoretical work [3]–[5] has shown that single-timestep ( $T=1$ ) SNN inference is approximately equivalent to INT8 quantized ANN inference: when the membrane potential is initialized to zero, the threshold-and-fire mechanism reduces to a ReLU-like clamp. Any NPU executing INT8 `Gemm+Relu` can therefore run an SNN-equivalent model.

We validate this on the STM32N6570-DK (ARM Cortex-M55 @ 800 MHz, Neural-ART NPU, 600 GOPS INT8). We focus on classifier inference latency, energy, and memory footprint on pre-extracted flow-level features; the full IDS pipeline (packet capture, flow aggregation) is left to future work.

Our contributions:

TABLE I  
HARDWARE-DEPLOYED IDS COMPARISON. ENERGY IS PER INFERENCE.

| Work             | Platform       | Task        | Acc.                           | Lat.           | Energy                             | NPU        |
|------------------|----------------|-------------|--------------------------------|----------------|------------------------------------|------------|
| Ngo [6]          | MAX78000       | bin.        | 98.6%                          | —              | 18 mW*                             | CNN eng.   |
| Zahm [7]         | Akida          | multi       | 98.4%                          | —              | $\sim 1$ W*                        | Neurom.    |
| Chehade [1]      | STM32F7        | multi       | 96.6%                          | 31 ms          | 7.86 mJ                            | None       |
| Diab [2]         | RPi 3B+        | multi       | 95.3%                          | 27 ms          | $\sim 6.75$ mJ <sup>‡</sup>        | None       |
| <b>This work</b> | <b>STM32N6</b> | <b>5/10</b> | <b>78.9/64.8%</b> <sup>†</sup> | <b>0.29 ms</b> | <b>44 <math>\mu\text{J}</math></b> | <b>NPU</b> |

\*Power, not energy; latency not reported.

<sup>†</sup>NSL-KDD 5-class / UNSW-NB15 10-class.

<sup>‡</sup>Estimated from reported power and latency.

- To our knowledge, the first publicly documented IDS classifier deployment on an ARM Cortex-M NPU, achieving 0.29–0.46 ms inference at an estimated 44–69  $\mu\text{J}$ .
- Empirical validation of  $T=1$  SNN–ANN equivalence on commercial NPU silicon, with 99% prediction agreement between FP32 and INT8 models.
- An operator-level analysis showing that the `Floor` operator required by QCFS [3] falls back to CPU, adding 17.4% latency.

## II. RELATED WORK

Table I compares IDS deployments on physical hardware. Ngo et al. [6] deployed a 1,360-parameter MLP on the MAX78000, an AI-specialized MCU with a fixed CNN accelerator, achieving 98.57% on UNSW-NB15 (binary) at 18 mW. Zahm et al. [7] used the BrainChip Akida AKD1000 neuromorphic processor (98.4%,  $\sim 1$  W). Both use purpose-built AI/neuromorphic hardware. Chehade et al. [1] deployed a 1D-CNN on the STM32F746G (Cortex-M7, no NPU), achieving 96.59% accuracy on ISCX VPN-nonVPN but requiring 31 ms and 7.86 mJ per inference. Diab et al. [2] deployed LightGBM/CNN on Raspberry Pi 3B+ (27–300 ms, 250–275 mW). Our work targets a general-purpose MCU with an attached programmable NPU, achieving two orders of magnitude lower energy than CPU-only MCU deployments.

Several GPU-based SNN-IDS systems have also been proposed: convolutional SNNs [8], event-driven SNNs [9], and transformer-spiking hybrids [10] report 94–99% accuracy on NSL-KDD/UNSW-NB15 but use binary classification and have no path to edge deployment. Our multi-class formulation (5/10-class) and hardware deployment target a different design point.

### III. SYSTEM DESIGN

#### A. $T=1$ SNN-ANN Equivalence

A Leaky Integrate-and-Fire (LIF) neuron updates its membrane potential  $V[t] = \beta \cdot V[t-1] + \mathbf{W}\mathbf{x}[t]$  and fires when  $V[t] > \theta$ . At  $T=1$  with  $V[0]=0$ , the leak term vanishes and firing reduces to  $\Theta(\mathbf{W}\mathbf{x} - \theta)$ , which is equivalent to a quantized ReLU. Bu et al. [3] formalized this via QCFS; Jiang et al. [4] and Bu et al. [5] further showed that INT8 post-training quantization produces a discretization grid that closely approximates the threshold-and-fire operation. In practice, any NPU that runs INT8 Gemm+Relu is executing SNN-equivalent computation.

#### B. Neural-ART NPU

The STM32N6570-DK pairs an ARM Cortex-M55 at 800MHz with the Neural-ART NPU rated at 600GOPS for INT8 and 3TOPS/W efficiency [11]. The NPU supports Conv, Gemm, Relu, Add, Clip, and MaxPool in INT8; unsupported operators fall back to the Cortex-M55 CPU with automatic cache-maintenance overhead at each NPU $\leftrightarrow$ CPU handoff. The board includes 4.2MB internal SRAM and 128MB external NOR Flash.

#### C. Model Architecture

We use a four-layer MLP with BatchNorm and ReLU:  $\text{Lin}(d \rightarrow 256) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Lin}(256 \rightarrow 256) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Lin}(256 \rightarrow 128) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Lin}(128 \rightarrow C)$ , where  $d=41$  (NSL-KDD) or 34 (UNSW-NB15) and  $C=5$  or 10. BatchNorm is folded into the preceding Linear at ONNX export, yielding a graph of Gemm+Relu operators only ( $\sim 110\text{K}$  parameters). The architecture is deliberately shallow: deeper models could improve minority-class recall but risk introducing operators outside the NPU’s supported set. All hidden dimensions are powers of two (256, 128), avoiding a known Neural-ART compiler issue with prime-number channel counts [11].

#### D. Datasets and Training

**NSL-KDD** [12]: 125,973 train / 22,544 test, 41 features, 5-class. **UNSW-NB15** [13]: 175,341 train / 82,332 test, 34 features, 10-class. Training: Adam ( $\text{lr}=10^{-3}$ ), cosine annealing, 80 epochs, inverse-frequency class weighting, 10 seeds. Three categorical features are label-encoded; remaining features are z-score normalized using training-set statistics. FP32 models are quantized to INT8 via ONNX Runtime static PTQ (MinMax, per-tensor, 1,000 calibration samples). A 24-configuration ablation (Section IV-C) confirms that the choice of calibration method, granularity, and sample size changes accuracy by at most 0.82 pp.

#### E. NPU Compilation and Operator Mapping

Models are compiled for the STM32N6570-DK using ST Edge AI Developer Cloud v4.0.0 (atonn v1.1.3). Table II shows the operator-level mapping. All Gemm+Relu layers map to NPU hardware epochs. The QuantizeLinear/DequantizeLinear operators at the

TABLE II  
OPERATOR MAPPING TO NPU EPOCHS. ReLU INT8 ACHIEVES NEAR-100% NPU UTILIZATION; QCFS INT8 FORCES CPU FALLBACK AT EVERY FLOOR.

| ONNX Operator    | NPU Support | Epoch Type | Notes                     |
|------------------|-------------|------------|---------------------------|
| Gemm (INT8)      | ✓           | HW         | Fused weight + bias       |
| Relu             | ✓           | HW         | Fused with preceding Gemm |
| Clip             | ✓           | HW         | Used by QCFS              |
| Floor            | ✗           | SW (float) | CPU fallback, +0.08 ms    |
| QuantizeLinear   | ✓           | HW/Hyb     | At model boundary         |
| DequantizeLinear | ✓           | HW/Hyb     | At model boundary         |

TABLE III  
TEST ACCURACY (%), MEAN $\pm$ STD, 10 SEEDS). MLP IS THE ONLY MODEL ELIGIBLE FOR NPU ACCELERATION.

| Dataset   | Model        | Overall                          | Macro F1                         | NPU?    |
|-----------|--------------|----------------------------------|----------------------------------|---------|
| NSL-KDD   | ReLU MLP     | <b>78.86<math>\pm</math>1.32</b> | <b>59.20<math>\pm</math>2.80</b> | ✓       |
|           | QCFS MLP     | 77.82 $\pm$ 1.11                 | 57.29 $\pm$ 2.77                 | Partial |
|           | Rand. Forest | 73.84 $\pm$ 0.19                 | 47.13 $\pm$ 0.33                 | ✗       |
| UNSW-NB15 | ReLU MLP     | 64.75 $\pm$ 0.61                 | 40.29 $\pm$ 0.90                 | ✓       |
|           | Rand. Forest | <b>69.46<math>\pm</math>0.10</b> | <b>48.63<math>\pm</math>0.38</b> | ✗       |

model boundary may require CPU epochs depending on the input dimension.

For the ReLU INT8 path: NSL-KDD compiles to 5 HW + 1 Hyb + 2 SW epochs (0.46 ms); UNSW-NB15 compiles to 4 HW + 0 Hyb + 0 SW epochs (0.29 ms, 100% NPU). The QCFS INT8 path compiles to 13 HW + 1 Hyb + 14 SW epochs (0.54 ms) because each Floor triggers a CPU round-trip via Dequant $\rightarrow$ Floor $\rightarrow$ Quant.

### IV. EXPERIMENTS

#### A. Classification Results

Table III reports multi-seed results. On NSL-KDD, the ReLU MLP outperforms both QCFS (Wilcoxon  $p=0.037$ ) and tree baselines. On UNSW-NB15 (10-class), Random Forest achieves higher overall accuracy, but cannot be compiled for the NPU: ST Edge AI Core rejects TreeEnsembleClassifier with “NOT IMPLEMENTED.” The accuracy gap reflects the difficulty of 10-class classification with extreme imbalance (Worms: 44 samples), not a modeling limitation specific to NPU deployment.

Per-class analysis on NSL-KDD reveals that DoS (F1=87.3%) and Normal (82.6%) are well-classified, while R2L suffers from low recall (25.0%) — 64.4% of R2L samples are misclassified as Normal. U2R (200 test samples) scatters across Probe (50.9%) and Normal (24.0%). On UNSW-NB15, Generic achieves F1=97.9%, but minority classes collapse: Worms F1=9.0%, Analysis F1=2.6%. These patterns are characteristic of extreme class imbalance, not NPU-induced degradation — FP32 and INT8 models agree on 99% of predictions.

Fig. 1 visualizes these patterns. The class imbalance is severe: R2L has only 995 training samples (0.8% of the set) yet 2,754 test samples (12.2%); U2R has 52 training samples.

|       | Predicted |       |       |       |       |  |
|-------|-----------|-------|-------|-------|-------|--|
|       | Norm      | DoS   | Probe | R2L   | U2R   |  |
| Norm  | 96.5%     | 0.6%  | 2.1%  | 0.3%  | 0.5%  |  |
| DoS   | 17.2%     | 80.2% | 2.3%  | 0.0%  | 0.3%  |  |
| Probe | 21.3%     | 8.3%  | 70.3% | 0.0%  | 0.1%  |  |
| R2L   | 64.4%     | 0.0%  | 0.8%  | 25.0% | 9.8%  |  |
| U2R   | 24.0%     | 0.0%  | 50.9% | 6.6%  | 18.6% |  |

Fig. 1. NSL-KDD confusion matrix (% , 10-seed mean, ReLU MLP). R2L→Normal (64.4%) and U2R→Probe (50.9%) dominate errors.

TABLE IV

NPU BENCHMARK ON STM32N6570-DK. ENERGY ESTIMATED FROM AN5946 [14] (~150 mW AT NOMINAL FREQUENCY); FP32 CPU-ONLY POWER DIFFERS AND IS NOT ESTIMATED (—).

| Model     | Dataset   | Time (ms)   | Energy (μJ) | HW | Hyb | SW | Flash (KB) | RAM (KB) |
|-----------|-----------|-------------|-------------|----|-----|----|------------|----------|
| ReLU FP32 | NSL-KDD   | 1.24        | —           | 0  | 0   | 11 | 466.4      | 2.17     |
| ReLU INT8 | NSL-KDD   | 0.46        | 69          | 5  | 1   | 2  | 137.7      | 1.25     |
| ReLU FP32 | UNSW-NB15 | 1.23        | —           | 0  | 0   | 11 | 461.9      | 2.14     |
| ReLU INT8 | UNSW-NB15 | <b>0.29</b> | <b>44</b>   | 4  | 0   | 0  | 120.6      | 0.50     |
| QCFS INT8 | NSL-KDD   | 0.54        | 81          | 13 | 1   | 14 | 138.0      | 2.00     |

### B. Energy, Latency, and Memory

Table IV presents benchmark results from ST Edge AI Developer Cloud, which cycle-accurately simulates the STM32N6570-DK. INT8 NPU execution yields **2.7×** speedup on NSL-KDD and **4.2×** on UNSW-NB15 over the same model on the Cortex-M55 CPU. At an estimated 150 mW (ST AN5946 [14]), this translates to 44–69 μJ per inference — **114–179×** less than Chehade et al.’s 7.86 mJ on the STM32F7 [1]. UNSW-NB15 achieves 100% NPU execution (4 HW, 0 SW epochs); Flash shrinks 3.4–3.8× from FP32 to INT8.

The energy figure is derived from the reference workload in AN5946, not direct measurement of our model. On-board validation with an STLINK-V3PWR probe is left to future work.

### C. INT8 Quantization Robustness

A 24-configuration ablation (3 calibration methods × 2 granularities × 4 sample sizes) shows that INT8 accuracy deviates from FP32 by at most 0.82 pp across all configurations. Several INT8 configurations slightly *improve* over FP32, likely due to regularization effects of quantization noise.

Table V presents layer-wise activation comparison between FP32 and INT8 models on 1,000 NSL-KDD test samples. Hidden layers show moderate cosine similarity (0.65–0.68) due to per-neuron quantization noise, but the logit layer recovers to 0.978 and the overall prediction agreement is 99.0%. INT8 maps each activation to one of 256 discrete levels; the bounded per-neuron noise does not propagate into classification errors. From the  $T=1$  SNN perspective, the quantized model (INT8) closely reproduces the firing decisions of the FP32 model.

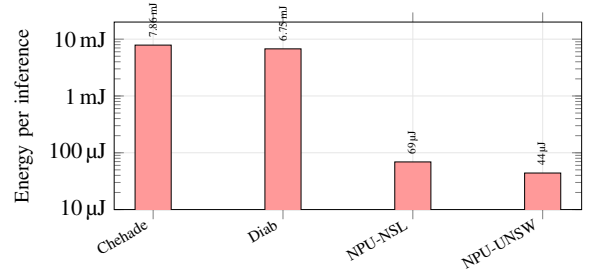


Fig. 2. Energy per inference (log scale). Our NPU path achieves over two orders of magnitude reduction versus CPU-only MCU baselines. NPU = ReLU INT8 on Neural-ART (estimated from AN5946). Chehade [1]: STM32F7; Diab [2]: RPi 3B+.

TABLE V

LAYER-WISE FP32 VS INT8 COMPARISON (1,000 NSL-KDD SAMPLES).

| Layer        | Cosine Sim   | MAE   | $L_{\infty}$ |
|--------------|--------------|-------|--------------|
| Relu_0 (256) | 0.667        | 0.435 | 43.7         |
| Relu_1 (256) | 0.655        | 0.438 | 62.6         |
| Relu_2 (128) | 0.683        | 0.400 | 65.2         |
| Logits (5)   | <b>0.978</b> | 0.103 | 19.8         |

## V. DISCUSSION AND CONCLUSION

**Design-space positioning.** This work targets a different point from GPU-based IDS: sub-millisecond, sub-millijoule inference on a commodity MCU. The absolute accuracy (78.9% NSL-KDD, 64.8% UNSW-NB15) is below GPU-simulation results, reflecting multi-class evaluation (5/10-class vs. binary) and NPU operator constraints (shallow MLP). Improving accuracy within these constraints is orthogonal to the deployment findings.

**Why NPU over CPU?** NPU inference frees the Cortex-M55 for RTOS tasks (packet capture, feature extraction), whereas CPU-only inference blocks the processor for 1.2 ms. Tree models achieve higher accuracy on UNSW-NB15 but have no execution path on the NPU (TreeEnsembleClassifier is not implemented in ST Edge AI Core).

**QCFS barrier.** The Floor operator is absent from the Neural-ART operator set, forcing CPU fallback with 17.4% latency overhead. ReLU INT8 remains the optimal activation for commodity MCU NPUs.

**RTOS implications.** At 0.29–0.46 ms, NPU inference fits within a typical 1 ms RTOS tick, and the Cortex-M55 remains free for concurrent tasks (Ethernet DMA, feature extraction) during NPU execution. The 1.2 ms CPU-only path would block the processor across tick boundaries. Validating this with μT-Kernel 3.0 is planned future work.

**Limitations.** We measure latency via ST Cloud benchmark; energy is estimated from the AN5946 application note, not direct measurement of our model. The evaluation covers classifier inference on pre-extracted features, not the complete IDS pipeline. NSL-KDD is a legacy dataset, and UNSW-NB15 has extreme class imbalance that limits absolute accuracy.

**Conclusion.** We demonstrated sub-millisecond, sub-millijoule IDS inference on a commodity MCU NPU, achieving  $2.7\text{--}4.2\times$  CPU speedup and over two orders of magnitude energy reduction versus CPU-only MCU deployments. Layer-wise analysis confirms 99% FP32/INT8 prediction agreement, validating  $T=1$  SNN-ANN equivalence on commercial NPU silicon. Code and trained models will be released upon acceptance.

#### REFERENCES

- [1] A. Chehade, E. Ragusa, P. Gastaldo, and R. Zunino, “Energy-efficient deep learning for traffic classification on microcontrollers,” in *Proc. IEEE ISCC*, 2025, arXiv:2506.10851.
- [2] A. Diab, A. Chehade, E. Ragusa, P. Gastaldo, R. Zunino, A. Baghdadi, and M. Rizk, “Intrusion detection on resource-constrained IoT devices with hardware-aware ML and DL,” *arXiv preprint arXiv:2512.02272*, 2025.
- [3] T. Bu, W. Fang, J. Ding, P. Dai, Z. Yu, and T. Huang, “Optimal ANN-SNN conversion for high-accuracy and ultra-low-latency spiking neural networks,” in *Proc. ICLR*, 2022.
- [4] H. Jiang, S. Anumasa, G. De Masi, H. Xiong, and B. Gu, “A unified optimization framework of ANN-SNN conversion: Towards optimal mapping from activation values to firing rates,” in *Proc. ICML*, ser. PMLR, vol. 202, 2023.
- [5] T. Bu, M. Li, and Z. Yu, “Inference-scale complexity in ANN-SNN conversion for high-performance and low-power applications,” in *Proc. IEEE/CVF CVPR*, 2025.
- [6] D.-M. Ngo, D. Lightbody, A. Temko, C. C. Murphy, and E. Popovici, “HH-NIDS: Hardware heterogeneous network intrusion detection system using MAX78000,” *Future Internet*, vol. 15, no. 1, p. 9, 2022.
- [7] W. Zahm, G. Nishibuchi, A. Jose, S. Chelian, and S. Vasan, “Low-power cybersecurity attack detection using deep learning on neuromorphic technologies,” CSIAC, Tech. Rep., 2024.
- [8] Z. Wang *et al.*, “An efficient intrusion detection model based on convolutional spiking neural network,” *Scientific Reports*, vol. 14, p. 7054, 2024.
- [9] S. N. Prajwalasimha *et al.*, “Event-driven intrusion detection systems using spiking neural networks for edge and IoT security,” in *Proc. IEEE ICSCSA*, 2025.
- [10] M. G. Karthik *et al.*, “Energy-efficient intrusion detection with a protocol-aware transformer-spiking hybrid model,” *Scientific Reports*, vol. 16, p. 7095, 2026.
- [11] STMicroelectronics, *ST Neural-ART NPU Concepts*, 2025, uM3225.
- [12] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, “A detailed analysis of the KDD CUP 99 data set,” in *Proc. IEEE CISDA*, 2009, pp. 1–6.
- [13] N. Moustafa and J. Slay, “UNSW-NB15: A comprehensive data set for network intrusion detection systems,” in *Proc. MilCIS*, 2015, pp. 1–6.
- [14] STMicroelectronics, *How to Optimize Low-Power Modes on STM32N6 MCUs*, 2025, application Note AN5946.